REPRESENTABILITY OF THE H^0 FUNCTOR OF A PSEUDO-COHERENT COMPLEX; PERFECTNESS UNDER GLOBALLY NONNEGATIVE FIBRES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma sites-cohomology-lemma-pseudo-coherent-pullback (canonical tag 08H4). cited at proof lines 686-687. Lg^*K is pseudo-coherent on T.
- Lemma spaces-perfect-lemma-pseudo-coherent (canonical tag 08JL). cited at proof lines 689-690 and 711-712. Cohomology sheaves of pseudo-coherent complexes are quasi-coherent; module translation.
- Proposition spaces-descent-proposition-fpqc-descent-quasi-coherent (canonical tag 04W8). cited at proof lines 691-692. Cech description of sections of quasi-coherent sheaves.
- Lemma bootstrap-lemma-locally-algebraic-space-finite-type (canonical tag 04U0). cited at proof line 701. Representability is etale local.
- Lemma spaces-morphisms-lemma-affine-local (canonical tag 03WG). cited at proof line 705. Affineness is etale local.
- Lemma spaces-morphisms-lemma-finite-presentation-local (canonical tag 0410). cited at proof lines 705-706. Finite presentation is etale local.
- Lemma spaces-perfect-lemma-derived-quasi-coherent-small-etale-site (canonical tag 071Q). cited at proof line 713. Zariski/module translation of K.
- Lemma spaces-perfect-lemma-descend-pseudo-coherent (canonical tag 08HE). cited at proof lines 713-714. Pseudo-coherence descends.
- Lemma perfect-lemma-pseudo-coherent (canonical tag 08E5). cited at proof lines 719-721. M^bullet represents K by a complex of A-modules.
- Lemma perfect-lemma-affine-compare-bounded (canonical tag 06Z0). cited at proof lines 720-721. Comparison of module and sheaf derived categories on affines.
- Lemma perfect-lemma-pseudo-coherent-affine (canonical tag 08E7). cited at proof lines 721-722. Realization over the affine base.
- Lemma more-algebra-lemma-cut-complex-in-two (canonical tag 068U). cited at proof lines 726-727. Canonical cut M^bullet = tau_{>=0} + tau_{<=-1}.

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- Lemma more-algebra-lemma-glue-perfect (canonical tag 066Y). cited at proof lines 733-734. Perfectness of the positive truncation.
- Lemma more-algebra-lemma-glue-tor-amplitude (canonical tag 066N). cited at proof lines 734-735. Tor amplitude [0, b] of the positive truncation.
- Lemma derived-lemma-negative-vanishing (canonical tag 05TC). cited at proof lines 742-743 (dual form). H^0(T, H^0(Lg^*K)) = H^0(T, Lg^*K).
- Definition more-algebra-definition-pseudo-coherent (canonical tag 064Q). cited at proof lines 751-752. Bounded-above finite-free representative of pseudo-coherent complexes.
- Lemma more-algebra-lemma-last-one-flat (canonical tag 0653). cited at proof lines 754-755. M = Coker(M^{-1} -> M^0) is flat.
- Lemma algebra-lemma-finite-projective (canonical tag 00NX). cited at proof lines 755-756. Flat finite presentation gives finite locally free.
- Lemma cohomology-lemma-bounded-flat-K-flat (canonical tag 06YD). cited at proof lines 761-762. The representative is K-flat, so derived pullback is term-wise.

Source reference. Original lemma 08H4: sources/stacks-chow/tags.txt, source lines 10767–10767. Cited at proof lines 686-687: Lg^*K is pseudo-coherent on T; load-bearing.

Source reference. Original lemma 08JL: sources/stacks-chow/tags.txt, source lines 10854–10854. Cited at proof lines 689-690 (cohomology sheaf quasi-coherent) and 711-712 (translation to modules); load-bearing twice.

Source reference. Original proposition 04W8: sources/stacks-chow/tags.txt, source lines 6351–6351. Cited at proof lines 691-692: the Cech description of H^0(T, F); load-bearing for the sheaf property.

Source reference. Original lemma 04U0: sources/stacks-chow/tags.txt, source lines 6273–6273. Cited at proof line 701: reduces representability to etale local on B; load-bearing.

Source reference. Original lemma 03WG: sources/stacks-chow/tags.txt, source lines 5111–5111. Cited at proof line 705: affineness checked etale locally; load-bearing.

Source reference. Original lemma 0410: sources/stacks-chow/tags.txt, source lines 5275–5275. Cited at proof lines 705-706: finite presentation checked etale locally; load-bearing.

Source reference. Original lemma 071Q: sources/stacks-chow/tags.txt, source lines 8999–8999. Cited at proof line 713: lets the proof think of K on the Zariski topology; load-bearing.

Source reference. Original lemma 08HE: sources/stacks-chow/tags.txt, source lines 10777–10777. Cited at proof lines 713-714: pseudo-coherence descends, completing the Zariski translation; load-bearing.

Source reference. Original lemma 08E5: sources/stacks-chow/tags.txt, source lines 10661–10661. Cited at proof lines 719-721: existence of the pseudo-coherent complex of A-modules M^bullet; load-bearing.

Source reference. Original lemma 06Z0: sources/stacks-chow/tags.txt, source lines 8904–8904. Cited at proof lines 720-721: K is the object corresponding to M^bullet; load-bearing.

Source reference. Original lemma 08E7: sources/stacks-chow/tags.txt, source lines 10663–10663. Cited at proof lines 721-722: M^bullet realizes K over the affine base; load-bearing.

Source reference. Original lemma 068U: sources/stacks-chow/tags.txt, source lines 8018–8018. Cited at proof lines 726-727: M^bullet = tau_{>=0} + tau_{<=-1} with the positive part perfect of tor amplitude [0, b]; load-bearing.

Source reference. Original lemma 066Y: sources/stacks-chow/tags.txt, source lines 7952–7952. Cited at proof lines 733-734: perfectness of tau_{>=0}M^bullet; load-bearing.

Source reference. Original lemma 066N: sources/stacks-chow/tags.txt, source lines 7942–7942. Cited at proof lines 734-735: tor amplitude [0, b] of tau_{>=0}M^bullet; load-bearing.

Source reference. Original lemma 05TC: sources/stacks-chow/tags.txt, source lines 7476–7476. Cited at proof lines 742-743 (dual form): H^0(T, H^0(Lg^*K)) = H^0(T, Lg^*K); load-bearing for replacing K by tau_{>=0}K.

Source reference. Original definition 064Q: sources/stacks-chow/tags.txt, source lines 7874–7874. Cited at proof lines 751-752 (with following discussion): M^bullet may be taken bounded above of finite free modules; load-bearing.

Source reference. Original lemma 0653: sources/stacks-chow/tags.txt, source lines 7887–7887. Cited at proof lines 754-755: M = Coker(M^{-1} -> M^0) is flat; load-bearing.

Source reference. Original lemma 00NX: sources/stacks-chow/tags.txt, source lines 875–875. Cited at proof lines 755-756: flat finite M is finite locally free; load-bearing.

Source reference. Original lemma 06YD: sources/stacks-chow/tags.txt, source lines 8882–8882. Cited at proof lines 761-762: the two-term-and-up complex is K-flat, so derived pullback is computed term-wise; load-bearing.

Lemma 0.2. *Let S be a scheme. Let B be an algebraic space over S. Let K be a pseudo-coherent object of $D(\mathcal{O}_B)$.*

(1) *If for all $g : T \to B$ in (Sch/B) the cohomology sheaf $H^{-1}(Lg^*K)$ is zero, then the functor*

$$(Sch/B)^{opp} \longrightarrow Sets, \quad (g : T \to B) \longmapsto H^0(T, H^0(Lg^*K))$$

is an algebraic space affine and of finite presentation over B.

(2) *If for all $g : T \to B$ in (Sch/B) the cohomology sheaves $H^i(Lg^*K)$ are zero for $i < 0$, then K is perfect, K locally has tor amplitude in $[0, b]$, and the functor*

$$(Sch/B)^{opp} \longrightarrow Sets, \quad (g : T \to B) \longmapsto H^0(T, Lg^*K)$$

is an algebraic space affine and of finite presentation over B.

Proof. Under the assumptions of (2) we have $H^0(T, Lg^*K) = H^0(T, H^0(Lg^*K))$. Let us prove that the rule $T \mapsto H^0(T, H^0(Lg^*K))$ satisfies the sheaf property for the fppf topology. To do this assume we have an fppf covering $\{h_i : T_i \to T\}$ of

a scheme $g : T \to B$ over B. Set $g_i = g \circ h_i$. Note that since h_i is flat, we have $Lh_i^* = h_i^*$ and h_i^* commutes with taking cohomology. Hence

$$H^0(T_i, H^0(Lg_i^*K)) = H^0(T_i, H^0(h_i^*Lg^*K)) = H^0(T, h_i^*H^0(Lg^*K))$$

Similarly for the pullback to $T_i \times_T T_j$. Since Lg^*K is a pseudo-coherent complex on T (Cohomology on Sites, Lemma 08H4) the cohomology sheaf $\mathcal{F} = H^0(Lg^*K)$ is quasi-coherent (Derived Categories of Spaces, Lemma 08JL). Hence by Descent on Spaces, Proposition 04W8 we see that

$$H^0(T, \mathcal{F}) = \mathrm{Ker}(\prod H^0(T_i, h_i^*\mathcal{F}) \to \prod H^0(T_i \times_T T_j, (T_i \times_T T_j \to T)^*\mathcal{F}))$$

In this way we see that the rules in (1) and (2) satisfy the sheaf property for fppf coverings. This means we may apply Bootstrap, Lemma 04U0 to see it suffices to prove the representability étale locally on B. Moreover, we may check whether the end result is affine and of finite presentation étale locally on B, see Morphisms of Spaces, Lemmas 03WG and 0410. Hence we may assume that B is an affine scheme.

Assume $B = \mathrm{Spec}(A)$ is an affine scheme. By the results of Derived Categories of Spaces, Lemmas 08JL, 071Q, and 08HE we deduce that in the rest of the proof we may think of K as a perfect object of the derived category of complexes of modules on B in the Zariski topology. By Derived Categories of Schemes, Lemmas 08E5, 06Z0, and 08E7 we can find a pseudo-coherent complex $M^\bullet$ of A-modules such that K is the corresponding object of $D(\mathcal{O}_B)$. Our assumption on pullbacks implies that $M^\bullet \otimes^{\mathbf{L}}_A \kappa(\mathfrak{p})$ has vanishing H^{-1} for all primes $\mathfrak{p} \subset A$. By More on Algebra, Lemma 068U we can write

$$M^\bullet = \tau_{\geq 0}M^\bullet \oplus \tau_{\leq -1}M^\bullet$$

with $\tau_{\geq 0}M^\bullet$ perfect with Tor amplitude in $[0, b]$ for some $b \geq 0$ (here we also have used More on Algebra, Lemmas 066Y and 066N). Note that in case (2) we also see that $\tau_{\leq -1}M^\bullet = 0$ in $D(A)$ whence $M^\bullet$ and K are perfect with tor amplitude in $[0, b]$. For any B-scheme $g : T \to B$ we have

$$H^0(T, H^0(Lg^*K)) = H^0(T, H^0(Lg^*\tau_{\geq 0}K))$$

(by the dual of Derived Categories, Lemma 05TC) hence we may replace K by $\tau_{\geq 0}K$ and correspondingly $M^\bullet$ by $\tau_{\geq 0}M^\bullet$. In other words, we may assume $M^\bullet$ has tor amplitude in $[0, b]$.

Assume $M^\bullet$ has tor amplitude in $[0, b]$. We may assume $M^\bullet$ is a bounded above complex of finite free A-modules (by our definition of pseudo-coherent complexes, see More on Algebra, Definition 064Q and the discussion following the definition). By More on Algebra, Lemma 0653 we see that $M = \mathrm{Coker}(M^{-1} \to M^0)$ is flat. By Algebra, Lemma 00NX we see that M is finite locally free. Hence $M^\bullet$ is quasi-isomorphic to

$$M \to M^1 \to M^2 \to \ldots \to M^d \to 0 \ldots$$

Note that this is a K-flat complex (Cohomology, Lemma 06YD), hence derived pullback of K via a morphism $T \to B$ is computed by the complex

$$g^*\widetilde{M} \to g^*\widetilde{M^1} \to \ldots$$

Thus it suffices to show that the functor

$$(g : T \to B) \longmapsto \mathrm{Ker}(\Gamma(T, g^*\widetilde{M}) \to \Gamma(T, g^*(\widetilde{M^1}))$$

is representable by an affine scheme of finite presentation over B.

We may still replace B by the members of an affine open covering in order to prove this last statement. Hence we may assume that M is finite free (recall that M^1 is finite free to begin with). Write $M = A^{\oplus n}$ and $M^1 = A^{\oplus m}$. Let the map $M \to M^1$ be given by the $m \times n$ matrix (a_{ij}) with coefficients in A. Then $\widetilde{M} = \mathcal{O}_B^{\oplus n}$ and $\widetilde{M^1} = \mathcal{O}_B^{\oplus m}$. Thus the functor above is equal to the functor

$$(g : T \to B) \longmapsto \{(f_1, \ldots, f_n) \in \Gamma(T, \mathcal{O}_T) \mid \sum g^\sharp(a_{ij}) f_i = 0,\ j = 1, \ldots, m\}$$

Clearly this is representable by the affine scheme

$$\operatorname{Spec}\Big(A[x_1, \ldots, x_n]/(\sum a_{ij} x_i; j = 1, \ldots, m)\Big)$$

and the lemma has been proved. □